

Overview

Note

What is being done: A rigorous critical audit evaluating the statistical weaknesses, biological assumptions, and computational constraints of Q5 Phase 4 phenotype characterisation and Ordinary Differential Equation (ODE) trajectory modelling.

Why we are doing it: To systematically document the methodological boundaries of our current patient stratification pipeline and identify vulnerabilities that could impact clinical interpretation.

What question it answers: What statistical, biological, and computational limitations restrict the predictive validity of Phase 4, and how can they be resolved in future refactoring iterations?

Phase 4 couples soft Gaussian Mixture Model (GMM) clustering of N=699 patients across four biological phenotypes (Immune Cold, Mutant-Driven, Immune Hot, Immunosuppressive M2-High) with dynamic 180-day ODE trajectory simulations (`ode_trajectories.png`). While this framework connects static multi-omic profiling to temporal treatment response, several critical statistical, biological, and data architecture limitations remain.

Statistical Limitations

Warning

What is being done: Evaluation of sample size asymmetries, multiple testing risks, and parameter approximation uncertainties within the Phase 4 analytical framework.

Why we are doing it: Unbalanced cluster distributions and unadjusted survival statistics can generate overconfident clinical conclusions for underrepresented patient subsets.

What question it answers: Where are the statistical findings most vulnerable to sampling noise, confounding, or parameter approximation errors?

Extreme Cluster Size Asymmetry and Low Subgroup Power

The Immune Cold cluster (Cluster 0) represents a tiny fraction of the pooled patient cohort, containing only N=22 patients (3.1% of N=699). When matched to the Q3 genomic parameter matrices for ODE trajectory simulation, this subpopulation shrinks further to just N=13 patients. Consequently, empirical response rates (50.0%) and mean biomarker signature Z-scores calculated for this group carry very wide confidence intervals, making statistical inference fragile and highly sensitive to sampling variation.

Unadjusted Survival Log-Rank Testing

Kaplan-Meier survival stratification across the four phenotypes (`km_survival_by_phenotype.png`) relies on unadjusted multivariate log-rank tests. These tests do not adjust for key clinical confounders, including patient age, prior systemic therapies, disease stage, or treatment cohort origin. Furthermore, because GMM clustering features (`TIS` , `CYT` , `CD8A` , `CD163`) partially overlap with established prognostic survival factors, there is an inherent risk of circularity in the survival evaluation.

Cluster-Level Multiplier Approximation (`pheno_r_mult`)

Although baseline proliferation rates (r) and immune killing coefficients (c) are parameterised per-patient via Q3 pERK coupling and checkpoint f_{kill} equations, the targeted therapy ODE arm relies on a cluster-level scalar (`pheno_r_mult` = 0.68 for Mutant-Driven, 1.25 for M2-High). Applying a single scalar multiplier across an entire phenotype cluster reintroduces a cluster-mean approximation, partially masking intra-cluster heterogeneity in drug response.

Literature-Derived Targeted Kinetic Rates

The kinetic rate constants governing tumour proliferation suppression under `BRAF` inhibition in the ODE targeted arm are literature-derived approximations. They have not been formally calibrated or fitted to the empirical cell-line drug viability AUC scores generated in Q2, limiting the numerical precision of simulated tumour regression rates.

Biological Assumptions and Model Boundaries

Warning

What is being done: Critical examination of biological mechanisms, cell types, and pharmacology missing from the current ODE trajectory formulation.

Why we are doing it: To clarify the biological boundaries of the differential equation model and prevent over-interpretation of simplified cell dynamics.

What question it answers: Which physiological tumour microenvironment features and drug resistance mechanisms are currently absent from the Phase 4 simulation?

Absence of an Explicit Stromal Compartment

The 2-state Kuznetsov ODE system models tumour cells (T) and immune effector cells (E), but lacks an explicit Cancer-Associated Fibroblast (CAF) or M2 macrophage compartment. Although M2-High tumours are defined transcriptomically by high CAF (`TGFβ1`) and M2 macrophage (`CD163` , `ARG1`) scores, the ODE represents stromal exclusion indirectly by lowering the killing coefficient (c) and elevating the proliferation multiplier (`pheno_r_mult` = 1.25), rather than modelling physical T-cell barrier mechanics through explicit cell-cell interaction equations.

Lack of Secondary Resistance and Phenotype Switching

Dynamic ODE simulations assume static drug sensitivity over the entire 180-day treatment period. The model does not incorporate acquired resistance mechanisms, such as secondary NRAS or MAP2K1 mutations, MEK bypass reactivation, or microenvironmental phenotype switching (e.g. an Immune Hot tumour transitioning to an Immune Cold state under anti- PDCD1 treatment pressure).

Drug Compound Mismatch Between Q2 and Phase 4

The targeted therapy arm of the ODE models Vemurafenib pharmacology based on Q3 Module A parameters ($K_D = 50\text{ nM}$). However, Q2 cell-line viability screens evaluated Dabrafenib and PLX-4720, but not Vemurafenib directly. Individual patient drug sensitivity AUC scores predicted in Q2 are not currently linked to the Phase 4 targeted therapy ODE, leaving an unresolved compound mismatch between the drug sensitivity and trajectory modules.

Targeted Arm Parameterisation Disconnect for NF1 -Loss Tumours

In the targeted therapy arm (Panel B of ode_trajectories.png), Mutant-Driven tumours are simulated under Vemurafenib BRAF inhibition, yielding partial regression ($T(180) = 0.79$) via pheno_r_mult = 0.68. However, empirical profiling reveals this cluster is 0% BRAF V600E mutant and 100% NF1 loss-of-function. NF1 loss eliminates RasGAP activity, resulting in high constitutive RAS-GTP levels. Under high RAS-GTP conditions, BRAF inhibitors induce RAF monomer-dimer transitions that paradoxically activate ERK signalling — the established RAF-inhibitor paradox. Simulating BRAF inhibitor sensitivity for NF1-loss tumours is mechanistically inaccurate; the biologically appropriate targeted agent for this cluster is MEK inhibition (MAPK1 / MAP2K1 targeting via Trametinib), which acts downstream of RAS.

Computational and Data Constraints

 **Warning**

What is being done: Audit of data integration paths, file dependencies, and cohort subset mismatches between Phase 4 and upstream modules.

Why we are doing it: To streamline computational data flows and eliminate architectural redundancies across project modules.

What question it answers: What technical data flow constraints exist between Q3 simulation outputs and Phase 4 reporting scripts?

Re-Simulated ODE Trajectories vs Direct Pre-Computed Ingestion

Phase 4 re-simulates dynamic trajectories locally by coupling Q3 steady-state pERK and checkpoint f_{kill} functions to 2-state Kuznetsov equations, rather than ingesting Q3's pre-computed 4-module patient simulation files (tumour_burden_simulations.csv). This creates architectural code duplication between q3-ode-model/ and q5-patient-stratification/.

Cohort Subset Discrepancy

While GMM soft clustering operates on the full pooled dataset of N=699 patients, dynamic ODE trajectory plotting in Phase 4 is restricted to the N=421 patients matched to Q3 expression and genomic parameter matrices (melanoma_params_full.csv). This creates a sample size discrepancy between baseline cluster characterisation (N=699) and trajectory plotting (N=421).

Actionable Fixes and Improvement Roadmap

 **Note**

What is being done: Presenting a prioritised, actionable roadmap to resolve identified statistical, biological, and computational limitations in future pipeline iterations.

Why we are doing it: To provide concrete engineering steps for enhancing model accuracy, clinical fidelity, and pipeline efficiency.

What question it answers: What specific modifications should be prioritised in the next refactoring cycle?

Priority	Proposed Improvement	Targeted Limitation	Expected Impact
P1 (Highest)	Direct Ingestion of Q3 Simulation CSVs	Re-Simulated ODE Trajectories	Eliminates code duplication by reading pre-computed Q3 tumour_burden_simulations.csv files directly into Phase 4 reporting.
P1	Re-parameterise Mutant-Driven Targeted Arm for MEK Inhibition	NF1 -Loss Targeted Disconnect	Replaces BRAF inhibitor simulation with Trametinib MEK inhibitor kinetics for NF1-loss tumours, resolving the RAF paradox error.
P1	Add an Explicit CAF / M2 Stromal ODE Compartment	Absence of Stromal Compartment	Introduces a 3rd differential equation for stromal density, enabling mechanistic simulation of physical T-cell exclusion in M2-High tumours.
P2	Calibrate Proliferation Rates to Empirical Q2 Viability Scores	Literature-Derived Targeted Rates	Fits targeted therapy r parameters directly to patient-specific Q2 Dabrafenib / PLX-4720 viability AUC predictions.
P2	Adjusted Cox Proportional Hazards and Multiple-Testing Correction	Unadjusted Survival Testing	Implements FDR-adjusted log-rank tests and multivariate Cox regression adjusted for age, stage, and cohort origin.
P3	Model Time-Varying Drug Efficacy $r(t)$ for Acquired Resistance	Lack of Secondary Resistance	Introduces time-decaying drug efficacy after Day 60 to simulate secondary NRAS mutations and MEK bypass resistance.

Key Takeaways

- Subgroup Power Deficit:** The Immune Cold phenotype is severely underpowered ($N = 22$ full cohort, $N = 13$ ODE subset), requiring cautious clinical interpretation of its empirical response rate.

- **Targeted Arm Disconnect:** Simulating `BRAF` inhibition on the Mutant-Driven cluster (100% `NF1` loss, 0% `BRAF` V600E) is biologically inaccurate due to the RAF paradox; future iterations must re-parameterise this arm for MEK inhibition (Trametinib).
- **Implicit Stromal Dynamics:** The 2-state ODE lacks a dedicated CAF compartment, relying on cluster-level multipliers (`pheno_r_mult = 1.25`) to approximate M2-High stromal exclusion rather than modelling cell-cell interaction physics.
- **Unlinked Q2 Viability Profiles:** Targeted therapy kinetic rates are literature approximations unlinked to Q2 per-patient drug sensitivity AUC predictions, leaving a compound mismatch between Vemurafenib and Dabrafenib.
- **Clear Refactoring Roadmap:** Prioritised improvements focusing on direct Q3 CSV ingestion, MEK inhibitor re-parameterisation, and 3-state stromal ODE expansion will significantly elevate Phase 4 model rigor.